

Azure Account Setup and Portal Familiarization

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Project Objective

The objective of this project was to create and configure a Microsoft Azure Free Tier account, explore the Azure Portal, create and manage resources, configure billing alerts, understand Azure security practices, and document the process with screenshots and explanations.

Microsoft Azure Practical Exercise Guide

1. Account Registration

Steps:

1. Open your web browser and visit:
<https://azure.microsoft.com/free>
2. Click **Start Free** or **Try Azure for Free**.
3. Sign in using an existing Microsoft account or create a new one.
4. Enter your personal details:
5. Full Name
6. Country/Region
7. Email Address
8. Verify your identity:
9. Enter your phone number.
10. Receive and enter the verification code sent by Microsoft.
11. Add a valid payment method (Debit/Credit Card).
12. Complete the registration process.
13. Sign in to the Azure Portal at:
<https://portal.azure.com>

Expected Result:

- Azure account created successfully.

- Free Trial subscription activated.
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2. Portal Exploration

Steps:

1. Log in to the Azure Portal.
2. Review the Home Dashboard.
3. Explore:
 4. Resource Groups
 5. Virtual Machines
 6. Storage Accounts
 7. Azure Active Directory
 8. Cost Management + Billing
9. Use the Search Bar at the top of the portal to locate services.
10. Pin commonly used services to your dashboard.

Expected Result:

- Familiarity with Azure Portal navigation.
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3. Governance Setup

Identify Subscription

1. Search for **Subscriptions**.
2. Open the Subscriptions service.
3. Locate your active subscription.

Create Resource Group

1. Search for **Resource Groups**.
2. Click **Create**.
3. Select your Subscription.
4. Enter Resource Group Name:
Example: RG-Lab01
5. Select a Region.
6. Click **Review + Create**.
7. Click **Create**.

Expected Result:

- Resource Group successfully created.
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4. Identity Awareness

Explore Azure Active Directory

1. Search for **Microsoft Entra ID** (formerly Azure AD).
2. Open the service.
3. Review:
 4. Users
 5. Groups
 6. Roles and Administrators

Review RBAC

1. Search for **Role-Based Access Control (RBAC)**.
2. Read Microsoft's RBAC documentation.
3. Understand common roles:
 4. Owner
 5. Contributor
 6. Reader

Expected Result:

- Basic understanding of identity and access management.
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5. Region Selection

Steps:

1. Visit:
<https://azure.microsoft.com/explore/global-infrastructure/geographies>
2. Review Azure Regions.

For users in Nigeria, recommended regions include:

- South Africa North (Johannesburg)

- South Africa West (Cape Town)
- Select the nearest region to reduce latency.

Expected Result:

- Region selected based on geographic proximity.
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6. Resource Deployment

Option A: Create a Storage Account

1. Search for **Storage Accounts**.
2. Click **Create**.
3. Select:
4. Subscription
5. Resource Group
6. Enter Storage Account Name.
7. Choose Region.
8. Leave default settings.
9. Click **Review + Create**.
10. Click **Create**.

OR

Option B: Create a Virtual Machine

1. Search for **Virtual Machines**.
2. Click **Create**.
3. Select:
4. Subscription
5. Resource Group
6. Enter VM Name.
7. Choose Region.
8. Select an Image:
9. Windows Server
10. Ubuntu Linux

11. Choose VM Size.
12. Configure Username and Password.
13. Review settings.
14. Click **Create**.

Expected Result:

- Resource successfully deployed.
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7. Cost Monitoring Configuration

Steps:

1. Search for **Cost Management + Billing**.
2. Open Cost Management.
3. Select your Subscription.
4. Open:
 - 5. Cost Analysis
 - 6. Budgets

Create Budget

1. Click Budgets.
2. Select Add.
3. Enter:
 - 4. Budget Name
 - 5. Amount
 - 6. Alert Threshold (e.g., 80%)
 - 7. Save.

Expected Result:

- Budget and cost monitoring configured.
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Required Evidence

1. Active Subscription Screenshot

Capture:

- Azure Portal Home Page
- Subscription Name
- Subscription Status

File Name:

Subscription-Screenshot.png

2. Resource Group Confirmation

Capture:

- Resource Groups Page
- Newly Created Resource Group

OR

Provide:

Resource Group Name: RG-Lab01

Region: South Africa North

Status: Active

3. Summary Report

Region Selected

South Africa North (Johannesburg)

Reason:

This region is geographically closest to Nigeria and typically provides lower network latency and better performance for hosted resources.

Shared Responsibility Model

For a Storage Account (PaaS):

Microsoft is responsible for:

- Physical infrastructure
- Networking
- Hardware maintenance
- Data center security

Customer is responsible for:

- Data protection
- Access control
- Account permissions
- Data classification

For a Virtual Machine (IaaS):

Microsoft manages:

- Physical hardware
- Data center
- Networking infrastructure

Customer manages:

- Operating System
- Applications
- User Accounts
- Security Patches
- Data

4. Cost Management Evidence

Capture:

- Cost Analysis page

OR

Capture:

- Budgets page showing budget configuration.

File Name:

Cost-Management-Screenshot.png